
Improving Search at Ounass

From Diagnosis to System Redesign

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CURRENT STATE

Current Search System

Retrieval

Keyword-based
on product name
& attributes

Ranking

Text relevance
Popularity
Business rules

Scale

Single-day dataset
4.1M impressions

From Impressions to Sessions

RAW DATA

- Each row = 1 product shown to 1 user
- No session identifier in the dataset
- 4.1M product impressions

4.1M

product impressions

CONSTRUCTED SESSIONS

- Session = user x keyword x query_id
- Detected via index-reset method
- Monotonic index; reset = new session

255K

unique query sessions: all analysis at session level

The State of Search

68.7%

Zero-Click Sessions

0.25

Overall MRR

0.86

MRR (Clicked Only)

Ranking works when retrieval works. The problem is retrieval.

Performance by Language

	Arabic	Latin
Share of queries	40%	60%
Catalog coverage	1%	99%
CTR	~2%	~2%
Conversion	~1%	~2%

Same engagement, half the conversion. 40% of demand hitting a 99% English catalog.

Performance by Intent

	Navigational (brand-specific)	Exploratory (By category/product)
Conversion	2.55%	1.32%
Success rate	27.2%	29.3%

Navigational queries show 2x higher conversion but lower success rate.

Navigational converts at 2x but fails more often: high-intent users blocked by retrieval.

Three Failure Patterns

Arabic Transliterations

- 69% failure : غوتشي (Gucci)
- 70% failure : شانيل (Chanel)
- Arabic keywords vs English catalog

Retrieval failure

Broad Categories

- "Evening gowns": 99.8% failure
- "Perfume": 91% failure
- Results returned, nothing relevant

Ranking / relevance failure

Long-tail Specific

- "D&G Slides in Rubber": 87%
- "Charlotte Tilbury Pillow Talk": 100%
- Multi-token fragmentation

Retrieval failure

SOLUTION DESIGN

The bottleneck is retrieval, not ranking.

Optimizations sequenced by complexity. Each addresses a specific gap.

Catalog Cleanup

~5%

catalog ghosts
with missing metadata

- Zero conversions -> remove from index
- Validate completeness at ingestion

Quick win — no engineering required

Incomplete products can't convert , so we remove them before they dilute relevance signals.

Arabic Transliteration Mapping

PHONETIC LOOKUP TABLE

Top 50–100 luxury brands
Arabic → English mapping

غوتشي

→ Gucci

شانيل

→ Chanel

كوتش

→ Coach

FUZZY FALLBACK

- Levenshtein distance matching
- Handles unlisted Arabic variants
- Edit-distance threshold tunable
- No manual curation needed for long tail

Addresses highest-failure navigational queries such as: Gucci 69%, Chanel 70%

Intent Router

NAVIGATIONAL

- Brand-specific query detected
- Route to precision retrieval
- Filter to brand → rank within

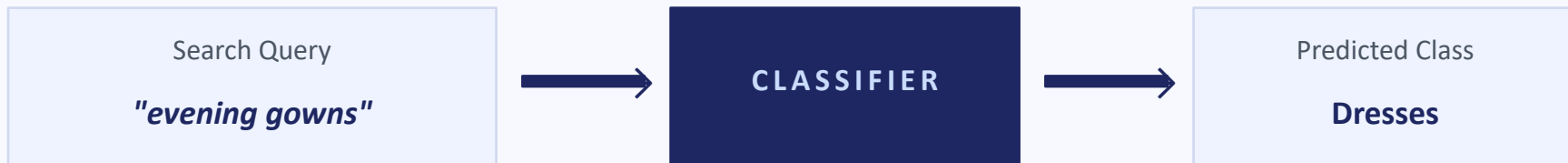
OR

EXPLORATORY

- Category / generic query detected
- Route to recall-oriented retrieval
- Broader candidate set + curation signals

Simple version is cheap to implement: substring match against known designers.

Query-to-Category Classifier



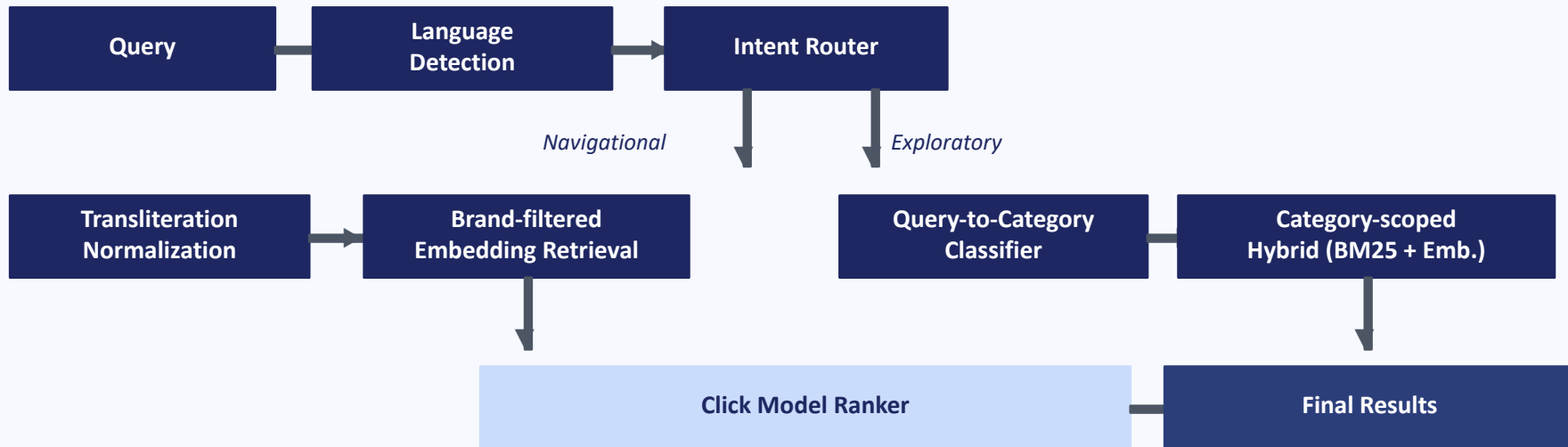
- 1 Constrains retrieval space before search executes
- 2 Reduces both false positives and false negatives
- 3 Enables efficient catalog indexing by product class
- 4 Composable with embeddings and BM25 downstream

Cross-lingual Embeddings

Model	Multilingual sentence transformers (e.g. paraphrase-multilingual-mpnet)
Semantic	Example: "evening gowns" retrieves "cocktail dress"
Cross-lingual	Arabic query غوتشي directly matches English product metadata
Hybrid	Combined BM25 + embedding retrieval: keyword precision + semantic recall

Highest impact, highest implementation cost

The Complete System



What About Ranking?

WHAT WE TESTED

- Logistic Regression, LightGBM, SVM
- 63 feature combinations (systematic ablation)
- Index debiasing method:
include-and-neutralize

WHAT WE FOUND

- All models converge $\rightarrow$ MRR = 0.2462
- Matches current system exactly
- Clean features can't reorder within a session
- Bottleneck is feature information,
not model capacity

The model needs query-dependent and semantically rich features which the retrieval redesign provides.

Evaluation: Offline

MRR

Mean Reciprocal Rank

How high is the first relevant result?

S@k

Success@k (k = 5, 10)

Is there a click in the top k?

Recall@k

Recall at k

How many relevant items retrieved out of total?

Offline evaluation confirms we haven't regressed. Only online evaluation confirms improvement since new retrieval surfaces products with no historical click labels.

Evaluation: Online

A/B TEST DESIGN

- User-level stratified randomization
- We can stratify by: language + activity tier
- Several weeks for significance
(luxury = sparse data)

METRICS

Primary

Revenue per search session

Conversion rate per session

Guardrails

Return rate

Average basket size

Secondary

CTR

Zero-click rate

Key Takeaways

- 1 Retrieval is the bottleneck: 68.7% zero-click rate, not ranking quality

- 2 Performance gaps are structural: language mismatch, intent-blind retrieval

- 3 Five optimizations, sequential by complexity: start with transliteration mapping alone

- 4 Ranking improves naturally once retrieval provides richer signals
